24CS320

USN											

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester BE: **Test-2**

Digital System Design

Duration: 1 Hr. 15 Min. **Date**: 08.1. 2026 **Max. Marks**: 30

NOTE: Question No. 1 is compulsory, answer either Question No. 2 or 3 completely.

Q.NO	CO	CD	PI	QUESTION	MARKS
1(a).	5	L2	2.3.1	Design mod 8 Synchronous Up Counter using JK Flip Flop with a neat diagram.	8
1(b).	4	L2	2.3.1	Explain SIPO and SISO shift registers with a neat diagram and truth table.	7
2(a).	2	L3	3.1.1	What are Multiplexers? Explain 4:1 MUX and 8:1 MUX with the necessary diagrams.	7
2(b).	3	L2	2.3.1	Explain Master Slave Flip flop using JK Flip flop with a neat timing diagram.	8
				OR	
3(a).	2	L3	3.1.1	Implement the following Boolean function using **8:1** MUX and **4:1** MUX for the following function: **$F(A,B,C,D)=\sum m(0,1,3,4,11,12,13,14,15)$**	7
3(b).	3	L2	2.3.1	Convert the following flip flops: i) SR flip flop to JK flip flop ii) SR to T flip flop to T flip flop	8

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Course Outcome: At the end of the course the students will have the ability to	
CO - 1	Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions.
CO - 2	Design and implement the combinational circuits.
CO - 3	Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops.
CO - 4	Illustrate the working of various registers and register circuits.
CO - 5	Design and implement synchronous/asynchronous counters and synchronous

Cognitive Domains	
L1	Knowledge
L2	Comprehension
L3	Application

PI's	
2.3.1	Able to apply computer engineering principles to formulate modules of a system with required applicability and performance.
3.1.1	Able to define a precise problem statement with objectives and scope.

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